

VAN REENEN, South Africa

WMO: 684710

Lat: 28.3789S

Lon: 29.3853E

Elev: 1680

StdP: 82.70

Time Zone: 2.00 (E02)

Period: 98-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	-2.1	-0.5	-13.4	1.4	13.3	-11.4	1.7	12.4	11.9	14.9	10.7	15.0	3.4	0	0.473

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	9.2	29.0	14.5	27.5	14.9	26.2	15.1	19.9	23.5	19.0	22.7	18.3	22.0	4.8	300

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.8	16.7	21.5	17.9	15.8	20.8	17.2	15.1	20.0	65.3	23.5	62.2	22.8	59.5	22.0	23.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
9.4	8.2	7.2	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
			DB												
			WB	-5.2	20.5	2.1	0.7	-6.7	21.0	-7.9	21.4	-9.1	21.8	-10.6	22.3

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	15.0	18.7	18.6	17.7	14.9	12.2	9.8	9.4	12.2	15.3	16.0	16.9	18.5		
	DBStd	4.55	2.56	2.28	2.48	2.86	2.77	2.96	3.56	3.54	4.17	4.03	3.56	3.16		
	HDD10.0	142	0	0	0	3	12	38	56	19	8	4	2	0		
	HDD18.3	1411	26	22	41	106	192	255	277	191	106	92	67	37		
	CDD10.0	1963	270	241	238	150	79	33	37	88	167	191	208	262		
	CDD18.3	189	38	29	20	3	0	0	0	1	13	21	23	41		
	CDH23.3	1315	201	112	99	27	3	0	1	16	166	219	175	298		
(13)	CDH26.7	231	34	8	5	1	0	0	0	0	27	46	33	76		
(14)	Wind	WSAvg	3.6	3.1	3.0	3.0	3.1	3.5	4.0	4.0	4.0	3.7	3.8	3.5		
(15) Precipitation	PrecAvg	695	136	97	78	35	12	6	6	15	21	69	84	119		
	PrecMax	982	292	166	124	102	67	25	46	69	74	180	141	227		
	PrecMin	330	52	38	15	1	0	0	0	0	0	7	19	25		
	PrecStd	163	62	41	26	21	16	7	12	17	19	49	40	47		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	30.0	28.1	27.5	26.1	23.3	21.1	22.0	25.3	29.1	30.2	29.9	31.5		
		MCWB	15.5	16.6	15.8	14.1	10.9	8.6	7.9	10.4	11.8	13.5	14.3	15.2		
	2%	DB	27.3	26.2	25.7	23.8	21.4	19.0	19.7	22.8	27.1	27.8	27.3	28.6		
		MCWB	17.4	17.3	16.1	14.0	10.4	7.9	7.4	10.0	11.4	13.3	14.6	15.6		
	5%	DB	25.6	24.7	24.3	22.0	19.9	17.3	17.9	21.0	25.2	25.9	25.3	26.4		
		MCWB	17.7	17.5	16.4	13.8	10.1	7.3	7.0	9.2	10.9	12.8	14.8	16.2		
	10%	DB	23.9	23.1	22.7	20.2	18.1	15.7	16.0	19.1	23.1	23.7	23.3	24.4		
		MCWB	17.5	17.3	16.3	13.4	9.7	7.0	6.5	8.3	10.4	13.0	14.8	16.4		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	21.3	20.6	20.2	17.5	13.5	10.6	10.0	13.0	14.7	17.2	19.6	19.8		
		MCDB	24.6	24.0	23.2	21.0	19.3	15.7	16.1	19.1	21.3	22.4	22.7	24.1		
	2%	WB	20.0	19.5	18.8	16.0	12.3	9.6	8.9	11.7	13.5	16.0	18.2	18.7		
		MCDB	23.9	22.9	22.4	20.1	17.5	15.0	14.8	18.5	20.4	21.3	21.8	23.0		
	5%	WB	19.0	18.7	17.9	15.1	11.4	8.9	8.0	10.8	12.6	15.2	17.0	17.8		
		MCDB	22.7	22.2	21.5	19.5	16.6	14.1	14.0	17.3	19.7	20.8	20.7	22.2		
	10%	WB	18.1	18.0	17.0	14.3	10.6	8.1	7.3	9.9	11.9	14.4	16.2	17.0		
		MCDB	21.7	21.5	20.6	18.4	15.9	12.8	13.3	16.1	18.9	19.7	20.3	21.5		
(35) Mean Daily Temperature Range	5% DB	MDBR	9.2	8.7	9.4	10.4	13.0	12.8	13.4	13.2	13.2	11.3	10.7	9.8		
		MCDBR	12.4	11.5	12.4	13.4	15.1	14.5	15.1	15.4	16.3	15.7	14.4	13.8		
		MCWBR	5.0	4.7	5.1	5.6	6.7	6.8	7.0	6.5	5.8	5.3	4.6	4.8		
	5% WB	MCDBR	10.1	9.6	10.0	10.6	12.4	11.8	13.1	13.0	13.1	12.0	10.9	11.0		
		MCWBR	5.6	5.3	5.5	5.4	6.4	6.7	6.6	5.9	5.7	5.4	5.5	5.3		
(40) Clear-Sky Solar Irradiance	taub		0.361	0.353	0.342	0.307	0.274	0.258	0.261	0.319	0.359	0.392	0.354	0.357		
	taud		2.388	2.416	2.421	2.486	2.555	2.565	2.537	2.370	2.278	2.231	2.373	2.385		
	Ebn at Noon		981	971	950	937	930	925	936	909	916	921	980	986		
	Edh at Noon		129	122	114	97	81	76	81	107	130	145	130	130		
(44) All-Sky Solar Radiation	RadAvg		6.48	6.15	5.43	4.55	4.01	3.65	3.93	4.61	5.47	5.76	6.44	6.66		
	RadStd		0.51	0.53	0.37	0.30	0.21	0.20	0.18	0.26	0.30	0.45	0.42	0.35		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB		1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (5 neighbors)	+0.57	+0.71	N/A	N/A	-0.39	-0.56	N/A	N/A	N/A	N/A

Nomenclature: See separate page